Lagrange's Equation:-

Lagrange's linear equation is $Pp + Qq = R$ where P, Q and R are functions of x, y and z.

To solve we use subsidiary equations

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}.$$

(Q1) Solve $(x^2 - y^2 - z^2)p + 2xyq = 2xz$

This is of the form $Pp + Qq = R$

where $P = x^2 - y^2 - z^2$, $Q = 2xy$, $R = 2xz$.

Consider $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$.

$$\frac{dx}{x^2 - y^2 - z^2} = \frac{dy}{2xy} = \frac{dz}{2xz}$$

Consider $\frac{dy}{2xy} = \frac{dz}{2xz} \Rightarrow \int \frac{dy}{y} = \int \frac{dz}{z}$

$$\log y = \log z + \log a$$

$$\log y = \log az$$

$$y = az$$

$$\Rightarrow a = \frac{y}{z} \rightarrow (1)$$

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Take multipliers x, y and z

we get $$\frac{x\,dx + y\,dy + z\,dz}{x(x^2-y^2-z^2)+y(2xy)+z(2xz)} = \frac{dz}{2xz}$$

$$\frac{2x\,dx + 2y\,dy + 2z\,dz}{x^3 - xy^2 - xz^2 + 2xy^2 + 2xz^2} = \frac{dz}{2xz}$$

$$\frac{2x\,dx + 2y\,dy + 2z\,dz}{x(x^2+y^2+z^2)} = \frac{dz}{xz}$$

$$\int \frac{2x\,dx + 2y\,dy + 2z\,dz}{x^2+y^2+z^2} = \int \frac{dz}{z}$$

$$\log(x^2+y^2+z^2) = \log z + \log b.$$

$$\log(x^2+y^2+z^2) = \log(zb)$$

$$x^2+y^2+z^2 = zb$$

$$\frac{x^2+y^2+z^2}{z} = b$$

Solution is $a = f(b)$ i.e $\frac{y}{z} = f\left(\frac{x^2+y^2+z^2}{z}\right)$.

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(Q9) Solve $(z-y)p + (x-z)q = y-x$.

This is of the form $Pp + Qq = R$

where $P = z-y$, $Q = x-z$, $R = y-x$

Form subsidiary equations $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\frac{dx}{z-y} = \frac{dy}{x-z} = \frac{dz}{y-x}$$

Use multipliers x, y, z,

$$\frac{x\,dx + y\,dy + z\,dz}{x(z-y) + y(x-z) + z(y-x)} = \text{each fraction}$$

$$\frac{x\,dx + y\,dy + z\,dz}{xz - xy + xy - zy + zy - xz} = \text{each fraction}$$

$$\Rightarrow \quad x\,dx + y\,dy + z\,dz = 0$$

$$\int x\,dx + y\,dy + z\,dz = a$$

$$\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = a$$

$$\frac{x^2 + y^2 + z^2}{2} = a$$

$$x^2 + y^2 + z^2 = 2a$$

$$x^2 + y^2 + z^2 = a_1$$

Use multipliers $1, 1, 1$

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$$\frac{dx+dy+dz}{(z-y)+(x-z)+(y-x)} = \text{each fraction}$$

$$\frac{dx+dy+dz}{0} = \text{each fraction} \Rightarrow dx+dy+dz = 0$$

$$\int dx+dy+dz = b$$

$$x+y+z = b.$$

Solution is $a_1 = f(b)$

$$x^2+y^2+z^2 = f(x+y+z)$$

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(Q3) Solve $(mz-ny)\frac{\partial z}{\partial x} + (nx-lz)\frac{\partial z}{\partial y} = ly-mx$

This is Lagrange's equation $Pp + Qq = R$

$P = mz-ny$, $Q = nx-lz$, $R = ly-mx$

The subsidiary equations are

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{mz-ny} = \frac{dy}{nx-lz} = \frac{dz}{ly-mx}$$

Choose multipliers x, y and z

Each fraction $= \dfrac{x\,dx + y\,dy + z\,dz}{x(mz-ny) + y(nx-lz) + z(ly-mx)}$

$$= \frac{x\,dx + y\,dy + z\,dz}{mzx - nxy + nxy - lzy + zly - zmx}$$

$$\Rightarrow x\,dx + y\,dy + z\,dz = 0$$

Integrating both sides

$$\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = c$$

$$x^2 + y^2 + z^2 = c_1 \rightarrow (1)$$

Choose multipliers l, m and n.

Each fraction $= \dfrac{l\,dx + m\,dy + n\,dz}{l(mz-ny) + m(nx-lz) + n(ly-mx)}$

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Each fraction $= \dfrac{l\,dx + m\,dy + n\,dz}{lmz - nyl + nxm - lzm + lyn - mxn}$

$\Rightarrow l\,dx + m\,dy + n\,dz = 0$

Integrating both sides

$lx + my + nz = C_2$

General solution is $F(C_1, C_2) = 0$

i.e, $F(x^2+y^2+z^2,\ lx + my + nz) = 0$

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